UNIT 11 CAPITAL MARKETS

Structure

11.0 OBJECTIVES

After going through this unit, you should be able to:

- Describe the purposes for which capital markets exist and the role they play in a modern economy;
- List the various participants and the types of exchanges taking place in these markets;
- Compare between debt and equity as means of raising finance for business firms;
- Analyse the working of debt markets, the instruments in these markets and the pricing in these markets;
- Discuss the principles of the working of equity markets; and
- Explain the concept of share price indices and the methods of their construction.

11.1 INTRODUCTION

The previous unit acquainted you with markets for short-term funds, and we learnt that these markets are called money markets. The present unit deals in detail with various markets for long-term funds. These are called capital markets. We will see there are primarily two types of capital markets: debt markets, and equity markets. In the next section, we consider the purpose of capital markets in general, and the various participants in these markets. In the section after that, we will look at debt and equity as alternative means of raising finance. Section 11.4 limits itself to the debt market but provides a detailed discussion about the workings of these markets. The subsequent section turns to equity markets and explains why there is volatility, or risk, in these markets. Finally, section 11.6 discusses the indices of share prices and how these are constructed. Let us begin by stating the purpose and pinpointing the uses of capital markets.

11.2 PURPOSE AND USES OF CAPITAL MARKETS

Investment is nothing but postponed consumption. So, there are investors who are willing to lock in their funds in the expectation of earning higher returns and making capital gains. On the other side, there are business firms, organisations, and individuals who need funds and finance for making physical investments in the hope of making profits or gains, or even for consumption. Capital markets constitute a mechanism for bringing together buyers and sellers of various types of financial assets. Exchanges of these financial assets determine their prices.

As we have discussed in previous units the financial system of a nation consists of financial institutions, financial markets, and financial instruments and services. A financial market is an arrangement that facilitates the exchange of financial assets, including deposits and loans, corporate stocks and bonds, government bonds, and derivatives like futures and options. You will read about derivatives in the next unit. Financial markets can be classified in several ways: as primary and secondary markets, that is markets in new issues and markets in existing issues; as organised or formal markets, and as unorganised or informal markets. Financial markets can also be classified as money markets, markets for short-term funds, and capital markets, or markets for long-run funds.

Capital markets can be markets where long-term debt contracts are traded, like bond markets or debentures, or these can be equity markets where financial claims are sold which allows the holder a share in the profits of the firms issuing these shares. The markets where shares are issued initially are called primary markets, and the markets where the holders of these shares further trade these in the expectation of making capital gains are called secondary markets.

Let us look now at the basis for the need and provision of finance, as this will enable us to know about debt as the foundation of finance, and see who the participants in the markets are. Capital markets are all about the raising of capital and matching those who want capital (borrowers) with those who have capital (lenders). One way of matching these are financial institutions like banks and other depository institutions. For more complex transactions, markets are needed in which borrowers (and their agents) can meet lenders (and their agents). Commitments to borrow and lend are created. There are also markets where these commitments to borrow and lend are themselves sold off to other people. For example, companies can raise money by selling shares to investors and existing shares can be freely bought and sold.

Who are the borrowers? Individuals can borrow for domestic purchases or longer term such as financing and mortgages of houses, consumer durables, or education... companies need short-term funds to finance their cash flow. They need longer-term funds for growth and expansion. Governments are huge borrowers. They often borrow by issuing securities like bonds. Their total borrowing is called National Debt. Public sector companies may also borrow for expansion and growth. Borrowing may also be done from foreign agencies, institutions etc. Similarly, lenders may be individuals, financial institutions, or even companies. A company can use some of its idle funds to lend for short periods to make gains, in the money market.

When money is lent, in many cases the borrower issues a receipt for the money, a promise to pay back. These receipts are called securities, and the lender has claims on these securities. Thus, securities are financial claims. The holders of financial securities are the lenders, and the issuers of financial securities are the borrowers. There are many types of securities like Treasury Bills, Certificates of Deposits, Commercial Paper, bonds, debentures, equity shares, and so on. All securities show key information about how much is owed, when it will be paid back, and how much is the rate of interest.

So, we have seen that markets exist for finance and the main participants are borrowers, lenders, their agents and other individuals and agencies that facilitate exchanges. Borrowers issue securities. Lenders purchase these securities. For example, when the government issues a bond, we say that the government is selling the bonds. When you purchase the bond, you are lending money to the government, and the government is borrowing money from you. The market for long-term securities is called a capital market.

Capital markets perform several important functions in a market economy. The capital markets make information processing and dissemination easier, which allows better decisions can be taken regarding investment, selling of assets, and so on. Capital markets allow quick valuation of financial instruments – both debt and equity. Capital markets also enable operational efficiency by simplifying transaction procedures and lowering search costs and transaction costs. Capital markets help to develop integration among several sectors of the economy such as between real and financial sectors, between debt and equity instruments, between short-term and long-term funds, and between domestic and external funds. Finally, capital markets facilitate the flow of funds into efficient channels through proper investment.

11.3 DEBT AND EQUITY AS MEANS OF RAISING FINANCE

In this section, we will take a look at one particular borrower: the business firm. We shall see how firms make decisions about raising finance for their short-term and long-term requirements. Firms can raise finance either by borrowing (debt) or by issuing shares. So when we are talking about shares, we have only the primary market in mind. We are looking from a firm's point of view and asking how much of the finance should be raised by incurring debt, and how much should be raised through the issue of shares. Is there some theory or some principles that will help the firm to make a decision? This is what we are discussing now.

A firm has several assets that it invests in, and these investments generate a stream of cash flows. If the firm is entirely equity financed, these cash flows belong only to the shareholders. On the other hand, the firm can offer shares but also go in for borrowings. This will offer the lenders some risk-free investment opportunities. How much of the capital the firm should raise through debt and how much through equity is a decision that is important for the firm. This is called the **capital structure** of the firm. To understand the decision regarding the capital structure of the firm, we will first look at some indicators of the firm's financial position, particularly some financial ratios. One can look at the balance sheet and the income statements. Balance sheets look at the situation with assets and liabilities at a particular point of time while the income statement is a view of the earnings and expenditure that takes place over a specified period, usually a year. We can also look at a firm's generation of cash. But what we propose to do now is to look at some important financial ratios regarding a firm's financial activity and performance. Ratio analysis forms a powerful tool for assessing the financial condition of the firm and enables us to relate disparate financial data with each other. Ratio analysis of this sort will enable us to understand the capital structure of the business firm.

Different people and groups have varying stakes in the firm, and they will have different concerns over a firm's performance. Stockholders and bondholders will be concerned about the long-term profitability of the firm, while creditors will look to see whether the firm can meet its payments on time. The firm's management is concerned with both the long run and short run as well as day-to-day performance. There are several types of ratios that we shall be considering now. All of these can be grouped into four types: **liquidity ratios**, which measure the ability and adequacy of the current assets to meet current liabilities as they come due; **activity ratios**, which indicate the efficiency with which the firm uses its resources; **financial leverage ratios**, which measure how effective the firm is in managing its debt; and finally, **profitability ratios**, which measure how effective the firm is in generating net revenues and income about sales, costs, assets and shareholder equity.

Let us begin with liquidity ratios. **Liquidity ratios** refer to a firm's ability to meet its short-term obligations and are concerned with the size and composition of the firm's working capital position. A higher working capital position implies more liquidity. An important ratio is the **current ratio,** which equals current assets divided by current liabilities. This ratio indicates the amount of current assets available to meet all its obligations under current liabilities. The current ratio includes inventories, but since inventories are among the least liquid of a company's assets, it is sometimes useful to exclude them. Deducting inventories from the current ratio gives us the **quick** or **acid-test ratio**:

$$\text{Quick ratio} = \frac{(\text{Current assets} - \text{Inventories})}{\text{Current Liabilities}}$$

Next, we turn to some activity ratios. These by and large have to do with sales generated, often about the amount invested in them. One such measure is the **accounts receivable turnover**, which equals net credit sales divided by turnover. A related measure is the average collection period, which is given by (365) (receivables)/ (net credit sales). We can look at the total asset turnover which is an overarching activity ratio and is defined as:

$$\text{Total asset turnover} = \frac{\text{Net sales}}{\text{total assets}}$$

Now let us focus on some financial leverage ratios, which will enable us to talk about the capital structure of firms. Sometimes, as in Britain, leverage is called 'gearing'. Leverage ratios reflect a firm's risk position by analysing its financing mix. The idea is that the more the operations of the firm are financed through debt, the greater the risk. Information on this can be obtained from the firm's balance sheet. The other way is to use income statement data to derive the coverage ratios, which measure the company's ability to service that debt. Two widely used ratios derived from the information in balance sheets are the debt ratio and the debt-equity ratio. By debt, here we mean the firm's current liabilities as well as long-run debt. The **debt ratio** is defined as:

Debt ratio = Total Liabilities divided by Total Assets

The **debt-equity ratio** is defined as:

Debt-equity ratio = Total Liabilities divided by Equities

Now since total assets equal total liabilities plus equity, it can be shown (it is left to you as an exercise) that:

Debt-equity ratio = Debt ratio divided by (1 – Debt Ratio)

From the income statements we can derive the firm's cash flow coverage, which is given as

$$\textbf{Cash Flow Coverage} = \frac{\text{EBIT} + \text{Lease/rental Payments} + \text{Depreciation}}{Interest + Lease/Rental\ Payment + Principal\ Repayment/\,(1-T)}$$

Where EBIT stands for annual earnings before interest and taxes, and T stands for taxes on a firm.

Finally, we discuss profitability ratios. These ratios allow the shareholders to evaluate management's performance. There are two types of profitability ratios used: profit margin on sales and returns on assets employed. Profit margins — a type of profitability ratio that looks at a firm's expenditure on sales. There are two important such ratios: gross profit margin and net profit margin. They are defined as follows:

$$\textbf{Gross Profit Margin} = \frac{\text{Gross Profits}}{\text{Net Sales}}$$

$$\textbf{Net Profit Margin} = \frac{\text{Net Income}}{\text{Net Sales}}$$

In addition, we can work out the **operating profit margin which is defined as:**

$$\textbf{Operating profit margin} = \frac{\text{EBIT}}{\text{Net Sales}}$$

In addition to looking at ratios related to the profit margin, we can relate returns to assets and shareholders' equity. Two ratios frequently used are **(ROA)**, also known as **return on investment (ROI)**, and **return on equity (ROE)**. They are defined as:

$$\text{Return on Assets} = \frac{\text{Net Income}}{\text{Total Assets}}$$

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

Equipped with these ratios, in the next section, we shall discuss, somewhat briefly, some elements of the capital structure of a firm.

Check Your Progress 1

1) What are the main functions performed by capital markets in the economy?

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2) Name and define two liquidity ratios and two financial leverage ratios.

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11.4 DEBT MARKET INSTRUMENTS AND THEIR PRICING

The main debt instruments are bonds, which are fixed-income securities. The investor (who is the lender) gets periodic interest payments over the life of the bond and the principal payment at the time of redemption of the bond. The issuer of the bond (who is the lender, that is who undertakes to bear the debt) promises to pay the holder (buyer) of the bond this periodic stream of payments and the principal at the end. Debentures are like bonds, except that there is a clause that enables the investor to convert the debenture into equity shares on certain pre-specified terms and conditions.

There are short-term debt instruments, which you studied in detail in the previous unit on money markets, like treasury bills, certificates of deposits, commercial paper etc. In this unit, we limit ourselves to a study of bonds. Bonds are issued by the government and these are government-guaranteed. For example, in India, the Government of India is the largest borrower. The government sells short-term to medium-term bonds issued by the Reserve Bank of India. State governments also sell bonds. Apart from the central and state governments, several government agencies issue bonds that are guaranteed by the government. Other than government bonds, there are bonds issued by public-sector companies, financial institutions, and private-sector companies.

Over the last couple of decades, there have been several innovations in the types of bonds that have been issued. Traditionally simple bonds have been issued (these are known as ‘plain vanilla bonds’ or straight bonds). These bonds usually pay a fixed periodic coupon over its life and return the principal on the maturity date. The new types of bonds are of several types: zero-coupon bonds; floating rate bonds; bonds with embedded options; and commodity-linked bonds. These are explained below:

Zero coupon bonds do not give regular interest payments. It is issued at a large discount and redeemed at face value on maturity. Sometimes, these bonds carry the call and put options. You will know about put and call options in the next unit. Floating rate bonds, unlike straight bonds that pay a fixed rate of interest, pay an interest rate that is linked to a benchmark rate such as the interest rate on Treasury bills. Bonds with embedded options include bonds like debentures that give the bondholder the right to convert them into equity shares; as well as callable bonds, which are bonds that give the issuer the right to redeem them prematurely on certain terms; and bonds with put options, which are bonds that give the investor the right to prematurely sell them back to the issuer on certain terms. Finally, there are commodity-linked bonds that are bonds the return from which depends on the price of certain specified commodities.

We will be looking at the risks associated with bonds in detail in unit 13 of Block 4. For the present, we proceed from the assumption that bonds are risk-free. With that assumption, let us now discuss the value of a bond and see how the value of a bond is determined. The value of a bond is equal to the present value of the expected cash flows arising from it. Here you will do well to go over unit 4 in Block 1. Hence we need to look at the expected stream of cash flows and an estimate of the returns. We assume further that the bond pays a fixed coupon interest rate, that coupon payments are made annually and that the bonds will be redeemed at par when it matures.

With these assumptions, the value for such a bond is:

$$A = \sum_{t=1}^{T} \frac{X}{(1+r)^t} + \frac{M}{(1+r)^T}$$

where A = value (in rupees)

T = number of years

X = annual coupon payment (in rupees)

The required return is denoted by r

M = maturity value

The period when the payment is received is denoted by t.

A basic property of a bond is that there is an inverse relationship between the price of the bond and yield. This is because as the yield increases, the present value of the cash flow decreases; therefore the price falls. What is the relationship between the bond price and time? Since the bond price must equal its par value at maturity (we assume there is no risk of default) bond prices change with time. If the current price prevails when the redemption is higher than the par value of the bond, the bond is said to be a premium bond; if the current price at the time of redemption is lower, the bond is said to be a discount bond.

The trading of bonds is usually based on their prices. But since there are significant differences in cash flow patterns, and also other features, of different bonds, different bonds are not usually compared in terms of their prices. Instead, the point of comparison is bond yields. We now turn to some measures of bond yields. A common measure of yield is the **current yield**, which relates the annual coupon interest to the market price. It is expressed as:

Current Yield = Annual Interest / Price.

The current yield does not consider the capital gain (or loss) that the investor will face if she holds the bond till maturity and sells at a premium (or discount). This measure also neglects the time value of money. The next measure of yield that we consider is the **yield to maturity**. This is the value of the rate of interest that makes the present value of the cash stream received from owning the bond equal to the price of the bond. It can be expressed as

$$P = \sum_{t=1}^{T} \frac{X}{(1+r)^t} + \frac{M}{(1+r)^T}$$

Where P is the price of the bond, and the other letters denote the same variable as in the previous equation. Sometimes the calculation of r in the above is difficult and a trial-and-error procedure is resorted to. The final measure of yield that we consider is the **yield to call**. It is calculated in the same way as yield to maturity, except that in yield to call, instead of maturity value at the end, there is call price and the duration, instead of going from say, 1 to T, goes from 1 to the number of years until the assumed call date.

Check Your Progress 2

1) What is yield to maturity?

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2) How do we calculate the expected stream of cash flows arising from bonds?

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11.5 EQUITY: MARKETS AND VOLATILITY

Before we go on to discuss equity markets, let us discuss the role that equity markets play in the economy. We shall use the terms stock market and stock exchange interchangeably. What does a stock exchange do? It provides the regulation of company listings, a price formation mechanism, the supervision of trading, authorisation of members, settlement of transactions and publication of trade data and prices.

Equity markets consist of primary markets and secondary markets. The primary market is the market for new issues of shares, where shares are initially offered to the public. The secondary market, on the other hand, is the market for already-issued financial assets. The main difference between the two markets is that in the secondary market, the issuer of the share does not receive funds from the buyer. Rather, the existing issue changes hands in the secondary market, and funds flow from the new buyer of the share to the original buyer.

Equity capital is ownership capital. It is the shareholders who collectively own the company. They undertake the risks but also earn the rewards of ownership. The amount of capital that a company *can* issue is the **authorised capital**. The amount of capital that a company offers to investors is the **issued capital**. The part of capital subscribed by investors is called the **paid-up capital** assuming that no calls are outstanding i.e., all the shareholders have paid their called-up money. The par value is the face value of the share. **Face value** is the equity share capital divided by the number of equity shares. When the issue price is greater than the par value, the difference is called the **equity premium**. The **book value** (sometimes called the intrinsic value) of an equity share is equal to:

(*paid-up capital + reserves and surplus*) /(*number of equity shares outstanding*). Alternatively, the book value can be found out by dividing the *net assets (assets minus liabilities)* by the number of equity shares outstanding. The **market value** of an equity share is the price at which it is traded in the market.

We now turn to some valuation ratios. Valuation ratios indicate how investors assess the stock in the equity markets. These are thus indicators of the performance of the issuing firm. The important valuation ratios are the price-earnings ratio, the yield, and the ratio of market value to book value.

Price-earnings ratio: It is defined as the market price per share divided by earnings per share. The earnings per share ratio is just the after-tax profit less preference dividend divided by the number of outstanding equity shares. In practice, this ratio is popular with the terminology PE ratio. Higher PE ratio shows higher earning to an equity share which may increase demand of the share.

The price-earnings ratio reflects risk characteristics, investor sentiments, the degree of liquidity, and growth prospects.

Yield: This measures the rate of return earned by shareholders. It is defined as:

$$\text{Yield } = \frac{\text{Dividend } + \text{ price change}}{\text{Initial price}}$$

We can break this into two parts:

$$\frac{\text{Dividend}}{\text{Initial price}} + \frac{\text{price change}}{\text{Initial price}}$$

The first term denotes the dividend yield, while the second term is the capital gains or losses yield. Usually, companies with low growth prospects offer a high dividend yield while the capital gains yield is low. It is the converse about companies with high growth prospects.

Market Value to Book Value ratio: This is defined as the ratio of market value per share to the book value per share. It is a measure of the firm's generation of wealth. If it is more than 1, that is, if the market value exceeds